

MARQUIS: A Three-Stage Pipeline for Video Retrieval-Augmented Generation

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Abstract

Retrieval-augmented generation from videos requires systems to retrieve relevant audiovisual evidence from large corpora and synthesize it into coherent, attributed text. Current approaches struggle at both ends: retrieval methods fail on complex, multi-faceted queries that cannot be captured by a single embedding, while generation methods lack the high-level reasoning needed to synthesize across multiple videos and face memory constraints over long, multi-video contexts. We present MARQUIS: a three-stage pipeline that addresses these limitations through (1) query expansion, fusion, and reranking, (2) calibrated structured evidence extraction, and (3) article generation from extracted evidence, optionally controlled by an RLM. On the MAGMaR2026 shared task, we improve retrieval performance from 0.195 to 0.759 (nDCG@10). For article generation, ITER-QA-BASE improves average human score from 3.09 to 3.83 over the CAG baseline, while MARQUIS-RLM achieves a human score of 3.30 and the strongest citation recall among non-QA systems.¹

1 Introduction

Large-scale video corpora now document real-world events with a breadth and immediacy that no single text source can match, yet turning this raw audiovisual evidence into a well-sourced analytical article remains largely a manual process. Grounded article generation (Martin et al., 2025a) from large video collections requires systems to retrieve relevant audiovisual evidence and synthesize it into coherent, attributable text.

Current video retrieval and generation systems each face distinct limitations. Retrieval methods struggle with complex information needs that combine multiple implicit and explicit sub-needs and

instructions in a single query (Weller et al., 2024), failing to surface all relevant videos to the information request. Generation methods face three interrelated challenges: long multi-video contexts exceed model memory constraints (Chen et al., 2024; He et al., 2024; Li et al., 2025), existing VLMs are not designed for multi-video reasoning, and most video understanding work remains focused on low-level recognition tasks like captioning and entity-centric QA rather than the high-level synthesis required for article generation (Martin et al., 2025a). These limitations compound in a full pipeline: retrieval errors propagate missing or irrelevant evidence into generation, where models already struggle to reason over the context they receive.

In this work, we present MARQUIS (Multimodal Article generation via Retrieval, Query decomposition, Uncertainty calibration, and Iterative evidence Synthesis), a three-stage pipeline that addresses these limitations through modular decomposition of retrieval, evidence extraction, and generation. First, we decompose and expand each query into sub-queries, retrieve independently over each sub-query, and fuse the resulting ranked lists before reranking. Second, we extract evidence from retrieved videos through complementary query-agnostic and query-conditioned components, then calibrate each extracted claim against its source video to estimate support probability. Third, we generate cited articles from the curated evidence, comparing single-prompt, clustering-based, and bullet-point strategies. We additionally introduce MARQUIS-RLM, an instantiation of Recursive Language Models (RLM; Zhang et al., 2026a) that treats each pipeline module as a callable tool within a persistent structured-memory environment, enabling iterative evidence gathering, cross-video conflict resolution, and fact curation before generating the final article. Our contributions can be summarized as follows:

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¹We release the code here: <https://github.com/debashishc/marquis>

1. We introduce MARQUIS, a three-stage pipeline for large-scale video retrieval-augmented article generation.
2. Our two-stage retrieval approach, combining query expansion with rank fusion and video-native reranking, improves nDCG@10 from 0.195 to 0.759 over a dense retrieval baseline on MAGMaR2026.
3. Our QA-based article generation approach, combining query decomposition with video-grounded question answering, improves average human score from 3.09 to 3.83 over the CAG baseline on MAGMaR2026 oracle article generation.

2 Related Work

2.1 Multimodal Retrieval and RAG

Retrieval-augmented generation (RAG; [Lewis et al., 2021](#)) grounds language model outputs in retrieved evidence. [Martin et al. \(2025a\)](#) formalize multi-video article generation, the task of retrieval-augmented generation from multiple videos.

Retrieval Retrieving videos has been widely studied, but [Kriz et al. \(2025\)](#) show that most methods are specialized to descriptive queries and do not generalize to semantic queries or scale to large corpora. However, bi-encoder methods for dense ([Luo et al., 2021](#); [Zhu et al., 2024](#); [Ma et al., 2025](#); [Li et al., 2026](#)), multi-vector ([Reddy et al., 2025](#); [Qin et al., 2026](#)), and modality fusion ([Samuel et al., 2025](#)) provide scalable options for retrieval. Video reranking ([Li et al., 2026](#); [Skow et al., 2026](#)) helps balance performance and scalability further, reranking the outputs of first-stage bi-encoder methods.

Generation Most work generating text from videos focuses on low-level extraction and single-video settings such as captioning and question answering ([Xu et al., 2016](#); [Krishna et al., 2017](#); [Lei et al., 2018](#); [Yu et al., 2019](#); [Zhang et al., 2025](#)). [Martin et al. \(2025a\)](#) show that existing VLMs fixate on low-level visual features and fail at the high-level synthesis required for article generation.

MARQUIS differs by integrating retrieval, calibrated extracted claims, QA-based evidence extraction, and iterative evidence control into a single system.

2.2 Long-Context Video Understanding

Recent long-video models extend temporal range through long-context training ([Li et al., 2025](#)), memory augmentation ([He et al., 2024](#)), and hierarchical compression ([Chen et al., 2024](#)), but still face computational and reasoning limits over extended multimodal context. Additionally, none of these methods are trained for multi-video settings. Recursive Language Models (RLMs; [Zhang et al., 2026a](#)) externalize long inputs into an interactive environment that can be inspected and processed through programmatic operations. We use this idea as a control layer for article generation: rather than placing all extracted evidence into one context, MARQUIS-RLM iteratively gathers, stores, and curates extracted claims before writing.

3 Retrieval

Our retrieval pipeline operates in two stages. First, we decompose each query into atomic sub-queries, retrieve independently over each, and fuse the resulting ranked lists into a single candidate set. Second, we rerank the fused candidates using a video-native reranker. [Figure 1](#) illustrates the full pipeline.

3.1 First-Stage

Our first-stage method consists of three key components: (1) Query Decomposition and Expansion, (2) Dense Retrieval, and (3) Rank Fusion.

Query decomposition and expansion. We focus on queries that are long, instructional requests that combine a professional persona, domain background, and multi-faceted information need. Dense retrievers, however, are typically trained on short, single-intent queries, and encoding a complex request as a single vector collapses its many sub-needs into one point in embedding space. To bridge this gap, we decompose each query into N atomic sub-queries using an LLM, where each sub-query targets a single retrievable fact and is phrased as a concise search phrase.

Dense Retrieval. Both original queries and decomposed sub-queries are retrieved against an omnimodal index, which produces a ranked list of the top 1,000 candidates for each query.

Rank fusion. Given N ranked lists per query, we aggregate them into a single ranking. Let $\text{rank}(v, q_i)$ denote the rank of video v in the list produced by sub-query q_i , and let $s(v, q_i)$ denote

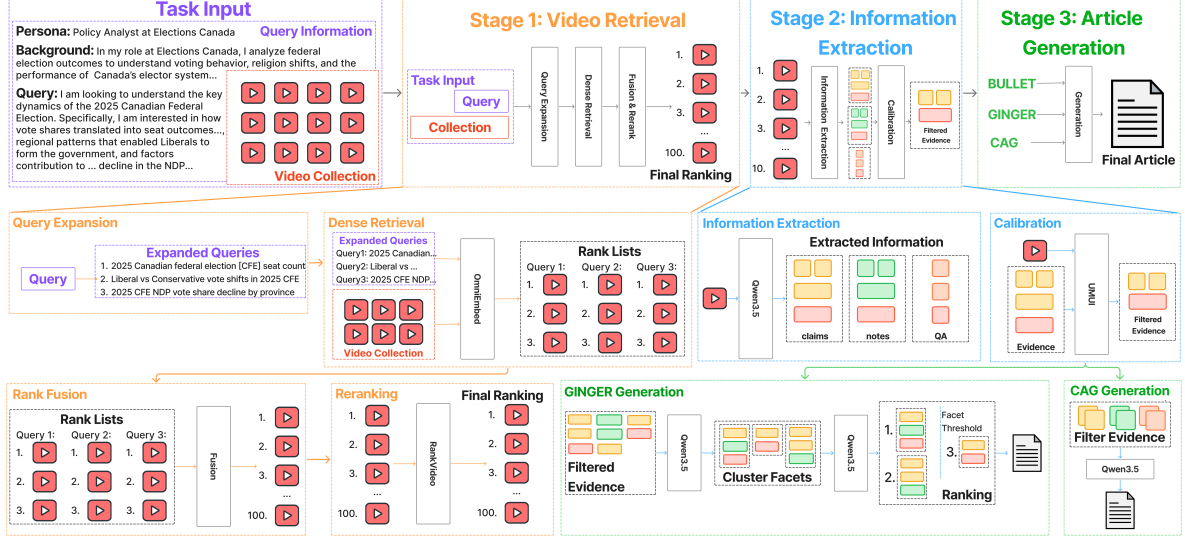


Figure 1: Overview of MARQUIS. **Stage 1 (Video Retrieval)**: Each query is decomposed into sub-queries, which are independently encoded by OmniEmbed and retrieved against the corpus. The resulting ranked lists are fused and reranked by RANKVIDEO to produce the final ranking. **Stage 2 (Information Extraction)**: Videos are processed by parallel information extraction streams—query-conditioned claims, query-agnostic notes, and QA—using Qwen3.5. Extracted evidence is scored by CLUE, a calibrated multimodal uncertain-inference model trained for the Unified Multimodal Uncertain Inference (UMUI) task, to filter unsupported claims. **Stage 3 (Article Generation)**: The filtered evidence is then passed to an article generator: BULLET passes the list of extracted claims as the article, CAG single-prompt baseline, or GINGER-based generation to produce a final cited article.

the cosine similarity score. We evaluate five fusion strategies:

- **Reciprocal Rank Fusion (RRF)**. Scores each video by the sum of reciprocal ranks across sub-query lists, with smoothing constant $K \in \{10, 60, 100\}$:

$$\text{RRF}_K(v) = \sum_{i=1}^N \frac{1}{K + \text{rank}(v, q_i)} \quad (1)$$

- **Sum of similarities**. Scores each video by the total cosine similarity across all sub-queries:

$$\text{Score}_{\text{sum}}(v) = \sum_{i=1}^N s(v, q_i) \quad (2)$$

- **Max similarity**. Scores each video by its highest similarity across sub-queries:

$$\text{Score}_{\text{max}}(v) = \max_i s(v, q_i) \quad (3)$$

- **Mean similarity**. Scores each video by the average similarity across sub-queries:

$$\text{Score}_{\text{mean}}(v) = \frac{1}{N} \sum_{i=1}^N s(v, q_i) \quad (4)$$

- **Weighted RRF**. Weights each reciprocal rank contribution by its cosine similarity:

$$\text{WRRF}_K(v) = \sum_{i=1}^N \frac{s(v, q_i)}{K + \text{rank}(v, q_i)} \quad (5)$$

Implementation details and expansion statistics are provided in [Appendix C](#).

3.2 Reranking

Given the top 100 videos per query from the first-stage retrieval, we perform reranking with RankVideo (Skow et al., 2026). For each full query, we pass the candidate videos from the first-stage retrieval to RankVideo and reorder the ranked list based on these judgments.

4 Information Extraction

The retrieval stage returns video candidates, but article generation requires finer-grained evidence. We therefore convert retrieved videos into extracted claims that can be selected, calibrated, cited, and passed to the generation stage. Our evidence extraction system contains three components: *query-agnostic* note extraction, *query-conditioned* claim extraction, and question-answer extraction. The

components differ in how they condition on the query, but all produce source-linked extracted evidence with video identifiers and, when available, timestamps. The extraction component is shown at a high level in [Figure 1](#); implementation details are provided in [Figure 2](#).

Let $V(q)$ denote the videos associated with query q . For each video $v \in V(q)$, the extraction stage may produce three evidence families:

$$N(v) = \{n_1, \dots, n_{|N(v)|}\},$$

$$C(v, q) = \{c_1, \dots, c_{|C(v, q)|}\},$$

$$A(v, q) = \{a_1, \dots, a_{|A(v, q)|}\},$$

where $N(v)$ denotes query-agnostic notes, $C(v, q)$ denotes query-conditioned claims, and $A(v, q)$ denotes question-answer outputs. Each output is later scored against its source video by the shared calibration stage.

4.1 Query-Agnostic Note Extraction

The query-agnostic component builds a reusable evidence base from each video without conditioning on a specific information need. Its goal is to capture directly observable visual events, on-screen text, and spoken content that may be useful across queries. The extractor is prompted to avoid speculation, causal inference, and cross-video synthesis. Each note describes a single atomic observation and includes a modality tag and optional timestamp. Confidence is not assigned at extraction time; support is estimated in a separate post-extraction calibration stage, which avoids conflating evidence extraction with support estimation.

4.2 Query-Conditioned Claim Extraction

The query-conditioned component targets evidence extraction toward a specific query. Where general notes prioritize breadth, this component prioritizes task relevance: given a specific information need, it extracts only claims that are relevant to that need. For each query-video pair, the extractor receives the query, persona, background, topic, and video metadata, and returns claims that are both query-relevant and directly supported by the video. This stage is not free-form answer generation: the prompt explicitly discourages generic scene descriptions, unsupported inferences, and redundant paraphrases. Each claim record contains a claim identifier, query identifier, video identifier, topic label, claim text, and optional support-oriented fields such as confidence,

evidence description, source type, and timestamp. The resulting claims provide a targeted evidence set for downstream article generation.

4.3 Question-Answer Extraction

The question-answering component extracts evidence through targeted video question answering for information needs. Given a query, persona, and background, the system decomposes the information need into atomic subqueries, retrieves relevant videos for each subquery, and uses a vision-language model to answer using the retrieved video content and transcript. We implement two variants: a single-shot variant that answers a fixed set of decomposed subqueries and aggregates the grounded per-video answers without introducing external knowledge, and an iterative variant that generates follow-up questions conditioned on previous question-answer history to pursue missing or underspecified information. The output is a set of question-answer evidence records, each linked to the question, answer, source video, and any available timestamp or confidence metadata.

4.4 Video-Grounded Calibration

After extraction, each output is scored against its source video. Given a video v and artifact $x \in N(v) \cup C(v, q) \cup A(v, q)$, calibration produces a support probability

$$s_\theta(v, x) \in [0, 1].$$

The score estimates whether the output is supported by the source video. Calibration is applied after extraction so that evidence creation and support estimation remain separate. The calibrated outputs retain their original text and metadata, with the support score attached for downstream filtering and article generation. Implementation details and prompts are provided in [Appendix E](#) and [Appendix I](#).

5 Article Generation

Our article generation pipeline synthesizes extracted video evidence into a fluent, source-attributed article that answers the information need. This stage operates after retrieval, evidence extraction, QA, and calibration. The article generator does not inspect raw videos directly, but instead consumes structured evidence artifacts tied to source videos and, when available, timestamps.

We design the article generation stage to be input-agnostic. In the experiments reported here,

the same generation procedures can operate over query-conditioned claims, query-agnostic notes, or question-answer pairs produced by the QA pipeline. Claims and notes include explicit video identifiers and timestamps, while QA pairs include the source videos used to produce the answer. The generator receives a flat list of extracted evidence together with their source metadata and is instructed to produce an article whose factual statements are supported by inline citations.

We compare three article generation strategies.

BULLET. The simplest strategy does not synthesize evidence into prose. It renders the retrieved evidence items directly as a numbered list of findings with inline citations. This output is conservative and preserves the connection between evidence and source videos, but it does not produce the coherent article-style response required by the task.

CAG is our Collaborative Article Generation baseline, adapted from WikiVideo (Martin et al., 2025a) to operate over extracted evidence. It generates a cited article from the extracted evidence for a query in one synthesis pass, following the role of the text-only aggregator in WikiVideo CAG. The model is instructed to organize evidence, remove redundancy, and preserve citations.

GINGER. We adapt the GINGER framework (Łajewska and Balog, 2025) to video-grounded extracted evidence. Since our extraction stage already produces atomic evidence, we skip nugget detection and perform facet clustering, cluster ranking, per-cluster summarization, and fluency enhancement. The model first groups evidence into thematic clusters (e.g., casualties, rescue efforts, government response), ranks them by query relevance, summarizes the top clusters independently into short cited sentences, and finally rewrites them into a coherent article. This staged-decision decomposes article generation into smaller controlled calls and helps preserve citations.

6 RLM Controller

In addition to the aforementioned pipeline, we further construct a RLM-based high-level control system for article generation, MARQUIS-RLM, organizing each module of the pipeline as a tool to be called by the Root LM.

Conceptually, RLM serves here as a general recursive control paradigm, whereas MARQUIS-RLM is our task-specific instantiation of this

paradigm for multi-video article generation. Unlike standard code-generating RLM, MARQUIS-RLM equips Root LM to call our pre-developed modules in REPL, preserving robust performance of specialized modules while gaining the reasoning and efficiency of the RLM paradigm. We further make explicit structured memory a core component of the system: Root LM always reasons over an evidence record that can be searched, reused, and revised, rather than relying only on what remains in context. This mitigates the information forgetting and multi-source confusion common in long iterative workflows, while making cross-video conflicts and missing information explicit. Examples are provided in [Appendix H](#).

REPL Environment and Tool Interface. We instantiate MARQUIS-RLM in a persistent Python sandbox whose namespace contains task context, memory bank, and callable sub-tools adapted from the modules in previous sections. At each iteration, the Root LM generates and executes code in the persistent namespace, accesses raw video, audio, and transcripts only through callable tools.

Memory Bank. Under MAGMaR’s long-context setting, the Root LM could face recurring failures including evidence forgetting, cross-source confusion, and missed conflicts or information gaps. All stem from the same limitation: the Root LM can only reason over what remains visible in context (Shi et al., 2026; Zhang et al., 2026c). Even the original free-form RLM-style REPL state is insufficient, since it introduces naming drift, re-assignment errors, schema drift, and perception-level confusion. To address this, inspired by recent external-memory designs for LM agents such as MemR³ (Du et al., 2025), we build a dynamic structured memory on top of the REPL.

The full schema and operators are given in [Appendix H](#).

Think-Act-Observe. We require the Root LM to follow a coarse-grained Think-Act-Observe mechanism, inspired by interleaved reasoning-and-acting frameworks such as ReAct (Yao et al., 2023).

This design enforces immediate external feedback at each step and grounds reasoning in explicit state transitions, while still leaving tool choice and reflection frequency to the Root LM.

	Run	nDCG@10	nDCG@20	nDCG@100	R@10	R@20	R@100
	OmniEmbed	0.195	0.229	0.311	0.190	0.276	0.494
<i>Sim</i>	Max Sim	0.722	0.743	0.784	0.639	0.731	0.826
	Mean Sim	0.637	0.650	0.696	0.544	0.618	0.736
	Sum Sim	0.703	0.725	0.776	0.604	0.698	0.818
<i>RRF</i>	K=10	0.700	0.739	0.777	0.612	0.735	0.832
	K=60	0.695	0.728	0.773	0.599	0.714	0.823
	K=100	0.688	0.719	0.767	0.590	0.704	0.818
	Weighted	0.699	0.730	0.778	0.604	0.714	0.832

Table 1: First-stage retrieval results. Best score per column is **bolded**.

7 Experiments

Dataset We evaluate on the MAGMaR2026 Test Set. The data is based on WikiVideo (Martin et al., 2025a). For the retrieval and RAG settings, we retrieve relevant videos from a combination of MAGMaR and MultiVENT2.0 (Kriz et al., 2025). For oracle article generation, systems receive the ground-truth relevant videos, isolating generation quality from retrieval quality.

Evaluation Setup Retrieval results are evaluated with nDCG and Recall for 10, 20, and 100. We use the ir-measures (MacAvaney et al., 2022) to calculate these scores. Generated articles are evaluated with an automatic and human evaluation. For the automatic evaluation, the systems are evaluated by MiRAGE (Martin et al., 2025b), which captures the factuality, information coverage, groundedness, and proper attribution of citations. Each MiRAGE entailment judgment is judged by CLUE (Zhang et al., 2026b). For human evaluation, three human annotators provide scalar scores of 1–5 for each system, scoring factuality, adequacy, coherence, relevancy, and fluency. After providing scalar scores for each prediction, the annotators also pick the best system response to each query.

Experimental Setup. All systems are evaluated under the same MAGMaR2026 retrieval and oracle-generation splits, but differ in their video access patterns and model backends. Retrieval uses OmniEmbed (Ma et al., 2025) for video and query encoding and Qwen3.5-9B (Team, 2026) for query decomposition. The information-extraction streams (note and claim extraction, calibration) use Qwen3.5-9B over sampled video frames. The QA pipeline combines Qwen3.5-27B for answer generation, Qwen2.5-Omni-7B (Xu et al., 2025) with Om-

niEmbed for multimodal embeddings, and Whisper medium.en (Radford et al., 2023) for transcription. Article generation uses Qwen3.5-27B for both the single-prompt baseline and GINGER-based generators. The RLM controller runs a Qwen3.5-9B root LM that calls the extraction and QA modules as sub-tools. None of the claim-based extraction or generation systems use audio; only the QA pipeline and RLM (via its transcription tool) access the audio stream. Frame rates, top- k values, generation budgets, and other component-specific hyperparameters are listed in Appendix A.

7.1 Retrieval

In Table 1 and Table 2, we report the results of video retrieval for first-stage and reranking, respectively. All query expansion and fusion methods substantially outperform the OmniEmbed dense retrieval baseline. This confirms that decomposing complex queries into sub-queries targeting atomic pieces of information is much more suitable for a dense retriever. This is an intuitive result, as most first-stage retrievers are trained on short, single-intent query-document pairs and compressing a complex information request into a single embedding is out-of-distribution and challenging (Weller et al., 2024). However, our sub-queries reduce this burden, allowing for the model to interface with in-distribution queries and leaving the merging of those ranked lists to a fusion or reranking approach.

Similarity vs. RRF fusion. Among first-stage methods (Table 1), Max similarity achieves the highest nDCG at all cutoffs. It benefits from its selection mechanism: because it scores each video by its best-matching sub-query, it surfaces videos that are highly relevant to at least one facet of the information need, even if they are irrelevant to oth-

Run		nDCG@10	nDCG@20	nDCG@100	R@10	R@20	R@100
OmniEmbed + RV		0.542 177.95	0.534 133.19	0.546 75.56	0.423 122.63	0.462 67.39	0.494 N/A
<i>Sim</i>	Max Sim + RV	0.399 -44.74	0.405 -45.49	0.425 -45.79	0.344 -46.17	0.383 -47.61	0.437 -47.09
	Mean Sim + RV	0.740 16.17	0.723 11.23	0.750 7.76	0.637 17.10	0.665 7.61	0.736 N/A
	Sum Sim + RV	0.747 6.26	0.758 4.55	0.800 3.09	0.636 5.30	0.711 1.86	0.818 N/A
<i>RRF</i>	RRF K=10 + RV	0.759 8.43	0.771 4.33	0.811 4.38	0.652 6.54	0.735 N/A	0.832 N/A
	RRF K=60 + RV	0.754 8.49	0.765 5.08	0.807 4.40	0.641 7.01	0.716 0.28	0.823 N/A
	RRF K=100 + RV	0.746 8.43	0.757 5.29	0.799 4.17	0.636 7.80	0.711 0.99	0.818 N/A
	Weighted RRF + RV	0.757 8.30	0.768 5.21	0.810 4.11	0.650 7.62	0.725 1.54	0.832 N/A

Table 2: Reranked retrieval results with percentage change relative to first-stage baseline. **Green** denotes improvement, **red** denotes degradation.

ers. RRF strategies (Cormack et al., 2009), which aggregate evidence across all sub-queries, produce more balanced rankings and achieve higher recall at deeper cutoffs, suggesting they are better at capturing the full breadth of a multi-faceted query. Mean and Sum similarity consistently underperforms the other aggregation methods, likely because averaging dilutes strong matches with weak ones. Among the RRF variants, lower K values perform slightly better, as a smaller smoothing constant amplifies rank differences and rewards videos that appear near the top of multiple sub-query lists. Weighted RRF performs comparably to standard RRF, indicating that weighting reciprocal ranks by cosine similarity provides limited additional signal when the sub-queries are already well-targeted.

Reranking. As shown in Table 2, applying RankVideo reranking improves performance across nearly all fusion strategies. Among the expanded-query methods, all RRF variants and similarity variants (except Max) see consistent improvements, with RRF at $K=10$ achieving the best ranking performance overall. The one notable exception is Max similarity, where reranking sharply degrades all metrics. We leave a detailed analysis of this failure mode to future work.

7.2 Generation

In Table 3, we report oracle generation results, where each system receives the ground-truth relevant videos rather than retrieved candidates, isolating generation quality from retrieval effects. We evaluate eight system variants spanning three evidence pipelines: claim-based extraction (BULLET, Ginger), question answering (iterative (Iter QA) and single shot (SS QA)), and RLM-controlled generation.

Generation Systems. Among the claim-based generation variants, GINGER is the strongest prose generator, improving over the CAG baseline in human score, best-vote share, and both information and citation precision. Its staged decomposition into facet clustering, ranking, and per-cluster summarization appears to help the model organize evidence and preserve citations more reliably than a single generation call. BULLET shows the opposite tradeoff: it achieves slightly higher citation recall than the other claim-based systems, but receives the lowest human score and no best-system votes, confirming that annotators penalize outputs that lack fluent synthesis even when source attribution is preserved. Taken together, these results suggest that explicit topical organization improves generation quality, but that the final output must still read as coherent prose to satisfy analyst information needs.

System	Human Score	Best Votes	Best %	Info		Cite	
				P	R	P	R
CAG (baseline)	3.09	1	1.8%	<u>76.4</u>	41.0	<u>61.7</u>	22.8
BULLET	2.67	0	0.0%	71.1	39.4	60.4	23.7
GINGER	3.12	6	10.5%	77.6	<u>40.4</u>	64.3	22.6
MARQUIS-RLM	3.30	3	5.3%	70.8	38.5	59.2	<u>27.2</u>
SS QA Base	3.07	6	10.5%	33.1	30.6	27.7	28.1
SS-QA-GINGER	3.42	10	17.5%	54.4	32.4	32.6	23.8
ITER-QA-BASE	3.83	<u>8</u>	<u>14.0%</u>	34.7	31.3	26.8	25.8
ITER-QA-GINGER	<u>3.69</u>	5	8.8%	34.5	29.0	25.7	22.6

Table 3: Oracle generation results for MARQUIS systems. H = human score; B = best-system votes; IP/IR = information precision/recall; CP/CR = citation precision/recall.

QA Systems. QA-based systems achieve the strongest human preference scores. ITER-QA-BASE obtains the highest average human score, while SS-QA-GINGER receives the most best-system votes. Their aggregate automatic metrics are weaker, largely because QA failures on a small number of topics produce conservative empty or near-empty outputs. This suggests that QA improves article usefulness when relevant answers are recovered, but remains brittle when decomposition or video-level answering fails. When the QA systems fail to answer questions, due to VLM failures or irrelevant sub-questions, the downstream generation systems often refuse to write the article, backing off due to insufficient evidence. This conservative behavior is a double-edged sword: it avoids hallucination on topics where the video evidence genuinely lacks the requested information (e.g., Myanmar Earthquake Q1), but it produces zero-score outputs that sharply deflate aggregate metrics on topics where information was available.

RLM. MARQUIS-RLM improves human score over CAG and BULLET and achieves the highest citation recall among non-QA systems. This suggests that iterative evidence gathering and structured memory help preserve attribution across multi-video contexts. Its lower precision and citation precision, however, indicates that the controller also admits less relevant facts into the final article. We therefore view MARQUIS-RLM as an evidence-management mechanism rather than a standalone replacement for structured generation; its Think–Act–Observe loop and persistent memory bank are effective at resolving

cross-video conflicts and filling information gaps (see examples in [Appendix H](#)), and tighter integration with GINGER-based synthesis is a natural next step.

8 Conclusion

We presented MARQUIS, a three-stage pipeline for video retrieval-augmented article generation. MARQUIS decomposes complex queries, retrieves and reranks relevant videos, converts video content into calibrated extracted evidence, and generates cited articles from selected evidence. The optional MARQUIS-RLM controller extends this pipeline by treating retrieval, extraction, QA, calibration, and generation as tools within a structured-memory environment, enabling iterative evidence gathering and curation before writing. Our experiments show that explicit query decomposition and video-native reranking substantially improve retrieval, while article-generation results reveal complementary tradeoffs among claim-based, QA-based, and RLM-controlled systems. More broadly, our findings suggest that grounded generation from video is best framed as an evidence-management problem. Rather than prompting a model to summarize long multi-video context directly, effective systems should retrieve broadly, extract atomically, estimate source support, and synthesize only from selected extracted evidence. Future work should improve learned calibration, integrate retrieval and extraction more tightly, and develop generation methods that combine structured evidence organization with iterative evidence control.

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A Experimental Setup Details

This appendix summarizes the model backends, input access, and hyperparameters used by each component. We report model backends and hyperparameters for retrieval, information extraction, calibration, and QA in [Table 4](#), and article generation hyperparameters in [Table 5](#). Prompt templates are listed in [Appendix I](#).

B Additional Results

[Tables 6, 7, 8, 9](#), and [10](#) report per-query scores across all systems and topics. Claim-based systems (CAG, Bullet, Ginger) consistently achieve high information precision but lower recall. QA systems suffer catastrophic zero-score failures on several topics, including the Alaskan Typhoon ([Table 6a](#)), Central Texas Floods ([Table 8a](#)), Nepal Youth Protests ([Table 9a](#)), and Shi Yongxin Scandal ([Table 10a](#)), where the QA pipeline fails to retrieve relevant videos and the generator backs off rather than hallucinate. The RLM performs most distinctively on the Canadian Federal Election ([Table 6b](#)), where cross-video conflict resolution yields the highest citation precision and recall on Q2, and on Nepal Youth Protests, where it substantially outperforms all other systems. The Palisades Fire ([Table 9b](#)) and Tropical Storm Wipha ([Table 10b](#)) exhibit uniformly high precision but very low recall across all systems, suggesting broad reference sets that no system fully covers.

In [Table 11](#) and [Table 12](#) we report the overall rankings of each system from the MAGMaR shared

Component	Backend	Setting	Value
Retrieval	OmniEmbed; RankVideo	Decomposition model	Qwen3.5-27B ($T=0.7$, $\text{top-}p=0.9$, 2048 tok)
		Thinking mode	Disabled
		Embedding pooling / norm	End-of-sequence; L2
		Precision	bfloat16
		Corpus size	109,814 videos
		First-stage depth	100 videos per (sub-)query
		Fusion methods	Max, Mean, Sum, RRF, WRRF ($K \in \{10, 60, 100\}$)
		Reranking depth	100 videos
Note / Claim Extraction	Qwen3.5-9B	FPS / max frames	1.0 / 128
		Decoding	$T=0.3$, $\text{top-}p=0.8$, $\text{top-}k=20$
		Max tokens (notes / claims)	2048 / 4096
		Seed (notes / claims)	42 / 40
		Thinking	off
Calibration	CLUE; prompted Qwen3.5	FPS	0.5
		Frame size	256×256
		Filtering threshold	0.5
QA	Qwen3.5-27B; OmniEmbed; Whisper medium.en	Questions / query	10–25
		Question decoding	$T=0.4$, $\text{top-}p=0.9$, 1024 tok
		Video QA / aggregation tokens	512 / 256
		Iterative max steps	5 / question
		Frame budget	32 frames
		Audio	16 kHz mono
RLM	Qwen3.5-9B (root, VLM); Qwen3.5-27B (judge); tools as above	Root LM context	32,768 tokens
		Caption VLM	32 frames, 32,000 tok
		Caption VLM decoding	$T=0.3$
		Max iterations	60
		LLM-as-a-Judge	$T=0.2$, 512 tok, per-iteration

Table 4: Unified component backends and hyperparameters for all MARQUIS pipeline stages.

Method	Input	Max Tokens	Decoding
BULLET	Selected claims	–	–
GINGER cluster	Claims	2048	$T=.3$, $p=.9$
GINGER rank	Clusters	512	$T=.3$, $p=.9$
GINGER summarize	Top 5 clusters	256	$T=.5$, $p=.9$
GINGER fluency	Summaries	1024	$T=.7$, $p=.9$

Table 5: Article generation hyperparameters. All methods use Qwen3.5-27B.

task leaderboards for retrieval and generation, respectively. Our retrieval systems hold the 2nd-6th place positions. Our generation systems place 1st and 3rd-6th.

C Appendix: Retrieval Implementation and Full Ablation

This appendix consolidates the retrieval implementation details, query expansion artifacts, retrieval figure, ablations, and hyperparameters. Prompt templates for query decomposition are listed in [Appendix I](#).

C.1 Query and Corpus Encoding

Each MAGMaR query is represented by concatenating the persona title, background, and query text. The original queries and generated sub-queries

are encoded with OmniEmbed using the same query prefix, appended end-of-text token, end-of-sequence pooling, and L2 normalization. The video corpus contains 109,814 videos, consisting of 109,724 MultiVENT 2.0 videos and 90 MAG-MaR2026 test videos. Search uses cosine similarity over normalized embeddings and returns the top 100 videos per query or sub-query.

C.2 Sub-query Expansion Statistics

The final flattened sub-query file contains 430 sub-queries across 19 original queries, for an average of 22.63 sub-queries per query. The minimum is 1 and the maximum is 25. The minimum is caused by one malformed decomposition output that fell back to a single query-like search probe in the flattened retrieval file. Excluding this fallback case, the 18

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	94.1	33.3	2.9	0.0
	Bullet	77.3	28.6	9.1	0.0
	Ginger	91.4	33.3	8.6	0.0
	RLM	81.5	28.6	3.7	0.0
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	19.4	3.2	0.0	0.0
	SS-G	27.3	4.8	2.3	0.0
Q2	CAG	87.5	41.3	4.2	0.0
	Bullet	94.7	34.9	0.0	0.0
	Ginger	87.5	41.3	0.0	0.0
	RLM	71.4	38.1	3.6	0.0
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	8.8	6.3	6.1	0.0
	SS-G	35.3	3.2	12.0	0.0

(a) 2025 Alaskan Typhoon

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	57.1	36.1	33.3	13.9
	Bullet	18.5	41.7	18.5	27.8
	Ginger	50.0	36.1	28.1	13.9
	RLM	27.3	44.4	21.2	50.0
	Iter-B	37.0	52.8	44.4	47.2
	Iter-G	22.2	44.4	27.8	47.2
	SS-B	31.2	38.9	28.6	38.9
	SS-G	38.5	33.3	38.5	38.9
Q2	CAG	43.4	61.1	42.3	33.3
	Bullet	46.7	41.7	46.7	13.9
	Ginger	43.6	61.1	43.6	33.3
	RLM	64.7	58.3	70.6	52.8
	Iter-B	43.4	61.1	43.4	58.3
	Iter-G	39.6	44.4	38.5	38.9
	SS-B	55.0	69.4	48.3	69.4
	SS-G	58.7	75.0	58.7	63.9

(b) 2025 Canadian Federal Election

Table 6: Per-query scores across all systems, judge: CLUE. Iter-B/G = Iter-QA-Base/Ginger, SS-B/G = SS-QA-Base/Ginger.

successfully decomposed queries produce 429 sub-queries, with a minimum of 22, an average of 23.83, and a maximum of 25 sub-queries per query.

C.3 Qualitative Expansion Examples

In Table 14 we show some example expansions from our method.

C.4 Fusion and Reranking

We evaluate both score-based and rank-based fusion over the sub-query ranked lists. The score-based methods are sum similarity, max similarity, and mean similarity. The rank-based methods are reciprocal rank fusion with $K \in \{10, 60, 100\}$ and weighted reciprocal rank fusion, where reciprocal-rank contributions are weighted by cosine similarity. We also evaluate reranked variants in which the top 100 first-stage candidates are reordered with RANKVIDEO.

C.5 Dropping Sub-queries

To test whether query decomposition depends on a small number of strong sub-queries or on broad facet coverage, we randomly retain only k sub-queries per original query before fusion, for $k \in \{1, 5, 10\}$. We repeat each random condi-

tion over five seeds and report mean and standard deviation. The full system uses all generated sub-queries.

Performance improves monotonically as more sub-queries are retained. A single random sub-query already outperforms the no-expansion baseline, showing that the decomposition often produces useful search probes. However, retaining 5 or 10 sub-queries substantially improves both ranking quality and recall, and using all sub-queries gives the best overall performance. This suggests that the gains from decomposition come not only from finding one strong reformulation but also from covering multiple facets of the original query.

D Appendix: Information Extraction Implementation

This appendix provides implementation details for the information extraction stage, including general note extraction, query-conditioned claim extraction, artifact schemas, representative outputs, and query–topic alignment. Prompt templates for note extraction, claim extraction, and calibration are listed in Appendix I. Figure 2 illustrates the information extraction and calibration workflow.

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	60.5	93.3	55.8	80.0
	Bullet	69.8	93.3	40.0	0.0
	Ginger	82.9	86.7	74.3	86.7
	RLM	68.4	60.0	54.1	46.7
	Iter-B	45.2	60.0	40.5	33.3
	Iter-G	38.3	60.0	27.7	46.7
	SS-B	17.6	40.0	15.7	40.0
	SS-G	30.2	53.3	14.0	60.0
Q2	CAG	78.1	93.3	63.3	80.0
	Bullet	61.9	86.7	71.4	80.0
	Ginger	76.5	93.3	60.6	80.0
	RLM	73.3	80.0	63.3	86.7
	Iter-B	68.4	86.7	59.6	66.7
	Iter-G	54.3	80.0	48.9	73.3
	SS-B	61.8	80.0	44.4	80.0
	SS-G	68.6	86.7	61.2	73.3

(a) 2025 Myanmar Earthquake

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	88.6	42.9	71.4	21.4
	Bullet	95.5	39.3	77.3	35.7
	Ginger	88.2	42.9	79.4	21.4
	RLM	63.0	42.9	55.6	42.9
	Iter-B	48.4	64.3	43.5	67.9
	Iter-G	63.5	71.4	57.7	46.4
	SS-B	62.9	67.9	44.9	64.3
	SS-G	72.1	67.9	63.4	67.9
Q2	CAG	77.4	57.1	67.7	17.9
	Bullet	72.2	57.1	55.6	46.4
	Ginger	80.8	57.1	69.2	17.9
	RLM	66.7	35.7	66.7	35.7
	Iter-B	30.3	71.4	22.7	71.4
	Iter-G	42.2	64.3	31.1	53.6
	SS-B	29.9	75.0	25.4	67.9
	SS-G	47.5	57.1	35.0	28.6

(b) Blue Ghost Mission 1

Table 7: Per-query scores across all systems, judge: CLUE. Iter-B/G = Iter-QA-Base/Ginger, SS-B/G = SS-QA-Base/Ginger.

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	65.5	20.0	61.1	11.1
	Bullet	56.2	15.6	56.2	13.3
	Ginger	66.7	20.0	71.4	6.7
	RLM	43.2	13.3	35.1	13.3
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	1.7	2.2	1.7	4.4
	SS-G	72.1	15.6	1.5	0.0

(a) Central Texas Floods

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	70.0	69.2	63.2	56.4
	Bullet	63.6	61.5	59.1	59.0
	Ginger	63.2	69.2	54.1	56.4
	RLM	44.8	61.5	41.4	38.5
	Iter-B	60.9	76.9	47.8	71.8
	Iter-G	75.0	76.9	66.7	69.2
	SS-B	64.6	74.4	64.6	76.9
	SS-G	75.9	64.1	67.9	59.0
Q2	CAG	72.4	48.7	69.0	43.6
	Bullet	64.3	51.3	57.1	48.7
	Ginger	67.7	48.7	71.0	43.6
	RLM	75.0	53.8	62.5	41.0
	Iter-B	51.6	53.8	39.1	56.4
	Iter-G	67.4	48.7	47.6	38.5
	SS-B	64.4	53.8	55.2	53.8
	SS-G	73.0	56.4	45.9	43.6

(b) Liberation Day Tariffs

Table 8: Per-query scores across all systems, judge: CLUE. Iter-B/G = Iter-QA-Base/Ginger, SS-B/G = SS-QA-Base/Ginger.

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	65.4	13.2	73.1	2.9
	Bullet	55.6	10.3	66.7	2.9
	Ginger	73.9	13.2	82.6	2.9
	RLM	88.6	29.4	94.3	17.6
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	13.0	2.9	6.7	1.5
	SS-G	35.7	11.8	3.6	0.0
Q2	CAG	95.2	32.4	100.0	7.4
	Bullet	89.5	38.2	84.2	23.5
	Ginger	92.9	32.4	100.0	7.4
	RLM	90.6	22.1	90.6	4.4
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	8.5	2.9	6.9	1.5
	SS-G	56.4	4.4	0.0	0.0

(a) Nepal Youth Protests

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	96.9	8.5	86.7	2.1
	Bullet	88.9	10.1	88.9	1.6
	Ginger	96.9	8.5	90.6	2.1
	RLM	93.3	16.9	86.7	8.5
	Iter-B	82.4	18.5	70.6	4.8
	Iter-G	75.0	12.7	69.4	6.3
	SS-B	56.0	18.0	54.0	11.6
	SS-G	58.0	13.8	52.0	5.8
Q2	CAG	97.7	7.9	95.2	2.1
	Bullet	94.4	9.0	100.0	3.2
	Ginger	98.0	7.9	93.8	2.1
	RLM	90.9	12.2	81.8	5.3
	Iter-B	80.3	13.8	78.7	10.6
	Iter-G	67.4	13.8	52.2	5.3
	SS-B	78.0	15.3	71.2	11.6
	SS-G	85.1	12.2	68.1	7.4

(b) Palisades Fire

Table 9: Per-query scores across all systems, judge: CLUE. Iter-B/G = Iter-QA-Base/Ginger, SS-B/G = SS-QA-Base/Ginger.

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	62.9	37.9	57.1	12.9
	Bullet	63.6	41.7	66.7	35.6
	Ginger	61.1	37.9	58.3	12.9
	RLM	70.0	46.2	70.0	28.0
	Iter-B	74.6	25.8	3.4	0.0
	Iter-G	80.8	24.2	7.7	0.0
	SS-B	17.9	9.8	10.3	3.0
	SS-G	64.8	26.5	57.4	0.0
Q2	CAG	73.0	37.9	70.3	18.9
	Bullet	76.0	43.2	84.0	35.6
	Ginger	76.5	37.9	70.6	18.9
	RLM	78.9	42.4	92.1	18.2
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	2.4	5.3	9.5	3.8
	SS-G	77.4	15.9	1.6	0.0

(a) Shi Yongxin Scandal

	System	Info F1		Cite F1	
		P	R	P	R
Q1	CAG	79.3	28.8	79.3	23.5
	Bullet	76.9	28.8	80.8	14.2
	Ginger	85.7	25.6	81.6	18.5
	RLM	85.7	22.8	68.6	17.4
	Iter-B	37.7	10.0	15.1	2.1
	Iter-G	29.5	9.3	13.6	3.2
	SS-B	32.3	8.9	29.0	3.9
	SS-G	45.7	7.1	25.0	3.6
Q2	CAG	86.2	15.3	75.9	5.0
	Bullet	84.6	15.7	84.6	8.2
	Ginger	90.3	15.3	83.3	5.0
	RLM	68.6	22.4	62.9	10.3
	Iter-B	0.0	0.0	0.0	0.0
	Iter-G	0.0	0.0	0.0	0.0
	SS-B	3.2	6.0	3.2	1.1
	SS-G	11.9	7.5	10.6	1.1

(b) Tropical Storm Wipha

Table 10: Per-query scores across all systems, judge: CLUE. Iter-B/G = Iter-QA-Base/Ginger, SS-B/G = SS-QA-Base/Ginger.

Rank	System	Avg.	nDCG@10	nDCG@20	nDCG@100	R@10	R@20	R@100
2	RRF K=10 + RV	0.759	0.759	0.771	0.811	0.652	0.735	0.832
3	Weighted RRF + RV	0.757	0.757	0.768	0.810	0.650	0.725	0.832
4	RRF K=60 + RV	0.751	0.754	0.765	0.807	0.641	0.716	0.823
5	Sum Sim + RV	0.745	0.747	0.758	0.800	0.636	0.711	0.818
6	RRF K=100 + RV	0.744	0.746	0.757	0.799	0.636	0.711	0.818

Table 11: MAGMaR retrieval final leaderboard positions for MARS submissions. Rank is the public leaderboard rank under the default average over the six displayed retrieval metrics.

Rank	System	Human	Best Votes	Best %	Info P	Info R	Cite P	Cite R
1	ITER-QA-BASE	3.833	8	14.0%	0.347	0.313	0.268	0.258
3	ITER-QA-GINGER	3.694	5	8.8%	0.345	0.290	0.257	0.226
4	SS-QA-GINGER	3.421	10	17.5%	0.544	0.324	0.326	0.238
5	MARQUIS-RLM	3.298	3	5.3%	0.708	0.385	0.592	0.272
6	GINGER	3.123	6	10.5%	0.776	0.404	0.643	0.226
8	SS-QA-BASE	3.070	6	10.5%	0.331	0.306	0.277	0.281
10	BULLET	2.667	0	0.0%	0.711	0.394	0.604	0.237

Table 12: MAGMaR oracle article-generation leaderboard snapshot for MARQUIS submissions. Rank is the public leaderboard rank under the default Human Score ordering.

D.1 General Note Extraction

General note extraction is run independently for each video. The extractor receives the video together with topic and video metadata and produces atomic observations describing directly observable visual content, OCR, and spoken or audio evidence. The output is a JSON object containing a list of notes. Each note includes note text, modality, and an optional timestamp.

D.2 Query-Conditioned Claim Extraction

Query-conditioned claim extraction is run for query–video pairs after aligning each evaluation query to a topic-specific video subset. The extractor receives the query identifier, topic, persona title, background, query text, and video identifier, and outputs claims relevant to the information need. Each claim is tied to a specific query and video and may include confidence, evidence description, source type, and timestamp metadata.

D.3 Query–Topic Alignment

The official query set contains 19 evaluation queries. Each query is aligned to one of 10 topic buckets through deterministic title-to-topic normalization. Query-conditioned claim extraction is then applied over the videos mapped to the corresponding topic. General note extraction uses the topic

identity only as metadata and does not condition on the evaluation query.

D.4 Extracted Evidence Schemas

A general note contains a note identifier, video identifier, topic label, note text, modality tag, and optional timestamp. A query-conditioned claim contains a claim identifier, query identifier, video identifier, topic label, claim text, and optional support-oriented metadata such as confidence, evidence description, source type, and timestamp.

D.5 Representative Outputs

To make the extraction flow concrete, we show one representative output from each major stage. These examples are lightly trimmed for presentation but preserve the actual field structure used by the pipeline.

Example general note.

```
{
  "note_id": "gn1a-hol6y3QwX2Y-000",
  "video_id": "hol6y3QwX2Y",
  "topic": "2025_Canadian_Federal_Election",
  "text": "A woman with short blonde hair and a beige jacket is speaking.",
  "modality": "visual",
  "timestamp": [0.0, 6.0]
}
```

Example query-conditioned claim.

Set	Queries	Total	Min	Avg.	Max
Final flattened retrieval file	19	430	1	22.63	25
Successful decompositions only	18	429	22	23.83	25

Table 13: Sub-query expansion statistics.

2025 Canadian federal election	2025 Alaska typhoon
2025 Canadian federal election final seat count by party	2025 Alaska typhoon housing damage assessment report
Elections Canada 2025 election official results dataset	residential structural failures Alaska 2025 typhoon
2025 Canadian federal election popular vote share by party	housing construction materials damaged 2025 Alaska storm
2025 Canadian federal election seat changes by party	geographic distribution of typhoon damage Alaska 2025
2025 Canadian federal election candidate vote totals	roof failure mechanisms 2025 Alaska coastal storm
voter turnout rate Canada 2025 vs 2021	foundation erosion damage coastal Alaska 2025

Table 14: Representative sub-queries produced by the query decomposition stage.

```
{
  "qc-10-1978302738418032640-000",
  "query_id": "10",
  "video_id": "1978302738418032640",
  "topic": "2025_Alaska_Typhoon",
  "claim": "More than 50 people have been rescued in Western Alaska.",
  "confidence": 0.95,
  "evidence": "Text overlay in the video states 'More than 50 people have been rescued in Western Alaska.'",
  "source": "video_text",
  "timestamp": [0.0, 3.0]
}
```

E Appendix: Calibration Implementation

This appendix provides implementation details for video-grounded calibration. Calibration is run after information extraction and assigns a support probability to each extracted artifact without modifying the original artifact content. The calibration prompt itself is provided in Appendix I.

E.1 Calibration Inputs and Outputs

For each extracted artifact, the calibration stage receives the source video and the artifact text. The output is a scalar support probability in $[0, 1]$ estimating whether the artifact is supported by the source video. The calibrated artifact preserves the original note or claim and attaches a calibration payload containing the support score and backend

provenance.

E.2 Backends

We evaluate two calibration backends. The prompted backend uses Qwen3.5 with a constrained probability-estimation prompt. The comparison backend is CLUE built on the Qwen2.5-Omni family. Both backends consume the same video-artifact pairs and emit the same conceptual output type.

E.3 Attachment Logic

Calibration predictions are attached to artifacts using stable artifact identifiers when available. When identifiers are unavailable or inconsistent, the attachment stage falls back to matching by video identifier and artifact text. This preserves compatibility across extraction and calibration jobs while keeping the original artifact representation unchanged.

Example calibrated artifact.

```
{
  "qc-10-1978302738418032640-000",
  "claim": "More than 50 people have been rescued in Western Alaska.",
  "calibration": {
    "unli": {
      "prob": 0.95,
      "raw": {
```

Sub-queries kept	nDCG@10	nDCG@100	R@10	R@100
1 random	0.613 ± 0.029	0.679 ± 0.025	0.512 ± 0.027	0.707 ± 0.024
5 random	0.684 ± 0.034	0.749 ± 0.029	0.601 ± 0.041	0.794 ± 0.031
10 random	0.696 ± 0.010	0.763 ± 0.013	0.615 ± 0.018	0.812 ± 0.015
All	0.711	0.773	0.640	0.831

Table 15: Effect of randomly retaining fewer sub-queries before max-similarity fusion. Random conditions are averaged over five seeds.

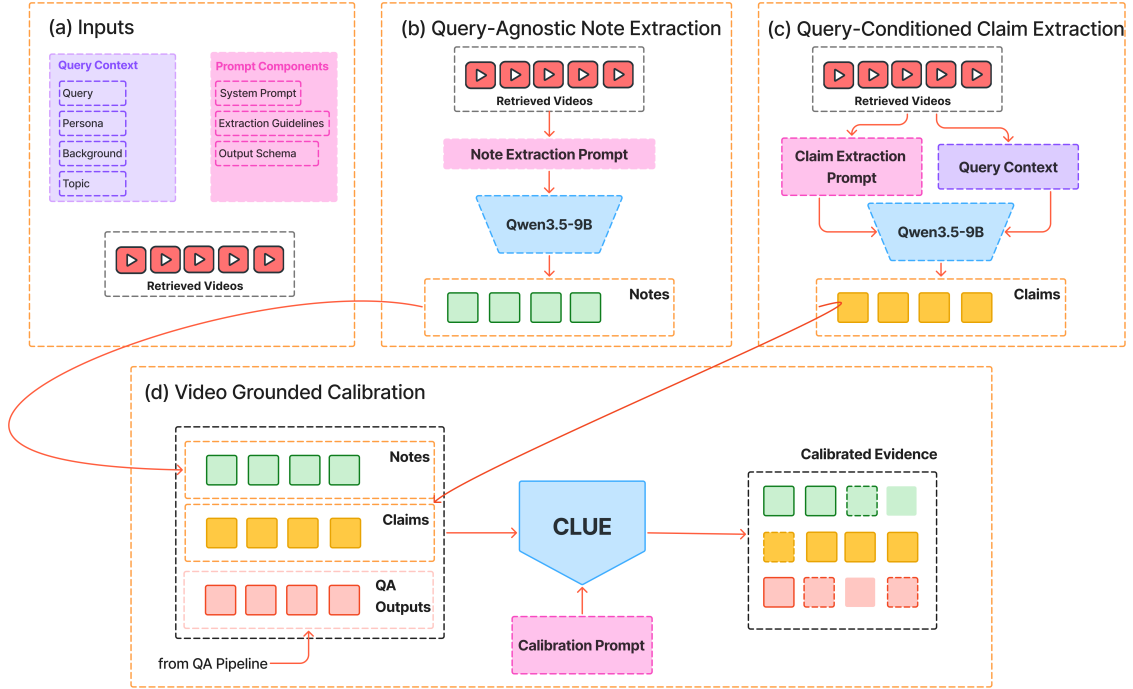


Figure 2: Information extraction and calibration workflow. Retrieved videos and prompt components are used to produce query-agnostic notes and query-conditioned claims, while QA outputs enter from the question-answering pipeline (See Appendix F and Figure 3). These extracted evidence records are merged and scored against source video by the calibration backend, producing calibrated extracted evidence for article generation.

```

    "raw_output": "<answer>0.95 </answer>"
  }
}
}

```

E.4 Claim Filtering

In addition to attaching support probabilities, the calibration stage can optionally filter extracted claims using the predicted support score. Claims with support probabilities below a predefined threshold are excluded from downstream outputs. This filtering mechanism is intended to reduce unsupported or weakly grounded claims while preserving high-confidence artifacts.

The filtering threshold is configurable and applied uniformly across calibration backends. Importantly, filtering is performed only after extraction and does not modify the original extracted content or calibration predictions.

For the calibration models (CLUE and Qwen3.5), the hyperparameters are summarized in Table 4.

F Appendix: Question Answering Implementation

Figure 3 provides an overview of the single shot and iterative question answering systems.

Single-Shot Question Answering. The single-shot pipeline uses Qwen3.5-27B for answer generation, Qwen2.5-Omni-7B with OmniEmbed-

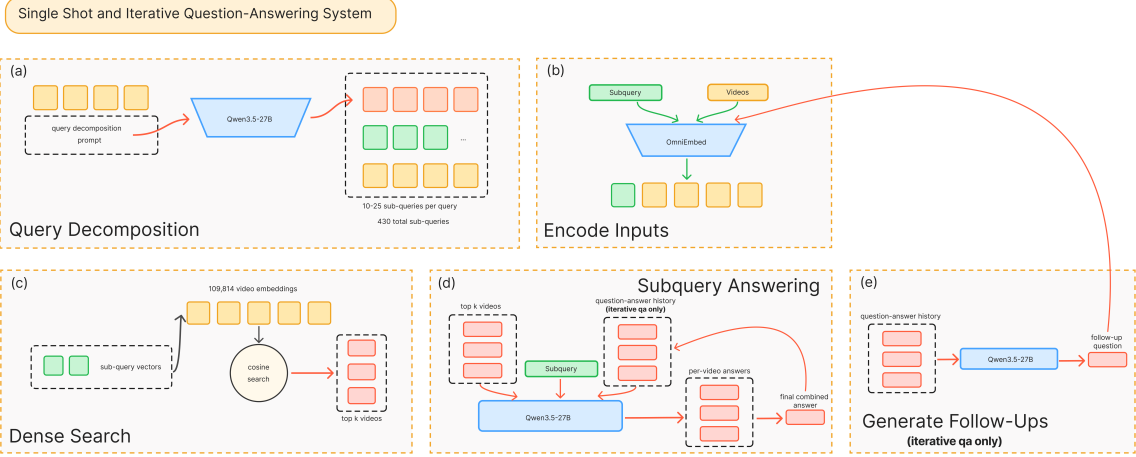


Figure 3: Overview of QA-based evidence extraction method. The single-shot variant decomposes the query into fixed subqueries, retrieves videos, answers each subquery, and aggregates the answers. The iterative variant generates follow-up questions from the question-answer history until a stopping condition is met.

v0.1 for multimodal embeddings, and Whisper (medium.en) for transcription.

Videos are preprocessed by downsampling to 30 FPS with a max frames of 32. Transcripts are generated from the audio stream and paired with each video. Video and query embeddings are computed in a shared space using the OmniEmbed model, and retrieval is performed using cosine similarity with a threshold of 0.1 and top-k selection with $k=4$.

For each sub-query, the system retrieves relevant videos and generates per-video answers using Qwen3.5-27B, conditioned jointly on the video frames, transcript, and query. The model is prompted to produce concise factual answers grounded only in the provided inputs. All per-video answers are collected, and responses such as “I don’t know” are filtered out. The remaining answers are then merged using a second Qwen3.5-27B pass that combines the extracted answers into a single response without introducing external knowledge.

Across the evaluated queries, the single-shot pipeline generated on average 23.84 expanded questions per main query, with a minimum of 20 and a maximum of 25 expanded questions. In the single-shot setting, 246 of 453 expanded questions were unanswered. Qualitative examples from the single-shot output for the first main query are shown below.

Iterative Question Generation. The iterative pipeline uses the same underlying structure as the single-shot approach, so each step performs re-

Expanded question	Answer
What were the final seat counts for each political party in the 2025 Canadian federal election?	Based on the provided video transcript with 96% of votes counted, the projected seat counts were Liberals: 166, Conservatives: 146, Bloc Québécois: 23, and NDP: 7. The final official counts after 100% of votes were counted were unknown based on the provided text.
What is the official dataset published by Elections Canada containing detailed results for the 2025 federal election?	I don’t know. The specific name of the official dataset published by Elections Canada containing detailed results for the 2025 federal election is unknown based on the provided information.
What were the popular vote percentages for the Liberal Party and Conservative Party in the 2025 Canadian federal election?	The Liberal Party received 43.1% of the popular vote and the Conservative Party received 41.7%.

Table 16: Examples of single-shot expanded questions and answers for the first main query.

trieval and per-video answer generation in the same way as the single-shot pipeline.

The key difference is that instead of processing each sub-query once, the system maintains a running history of question-answer pairs and iteratively refines the query. After each retrieval and answer aggregation step, the aggregated answer is appended to the history, and a new follow-up question is generated using Qwen3.5-27B with sam-

pling enabled, prompting the model to produce exactly one question that extracts additional or more specific information conditioned on the full history. This loop continues for up to 5 steps per sub-query but terminates early if no videos are retrieved, the model outputs “NONE” as the next question, or a repeated question is detected.

Across the evaluated queries, the iterative pipeline generated a minimum of 22 expanded questions, a maximum of 73 expanded questions, and an average of 41.05 expanded questions per main query. In the iterative setting, 293 of 613 expanded questions were unanswered.

Qualitative examples from the iterative output for the first main query are shown below. These examples illustrate how the iterative method can generate follow-up questions that become more specific than the original expanded questions.

Expanded question	Answer
What were the final seat counts for each political party in the 2025 Canadian federal election?	Based on the provided transcript with 96% of votes counted, the projected seat counts were Liberals: 166, Conservatives: 146, Bloc Québécois: 23, and NDP: 7. The final official seat counts were unknown because the provided data represented projections before 100% of votes were counted.
Which specific electoral districts accounted for the largest swing in votes that resulted in the projected reduction of the NDP’s seat count to seven?	Fortress Vancouver, Fortress Montreal, and the GTA accounted for the largest swing in votes that resulted in the projected reduction of the NDP’s seat count to seven.

Table 17: Examples of iterative expanded questions and answers for the first main query.

G Appendix: Article Generation Implementation

This appendix provides implementation details for the article generation systems. Prompt templates for all article generation variants are provided in Appendix I.

G.1 Evidence Inputs

The article generation systems operate over flat lists of evidence artifacts. These artifacts may be query-conditioned claims, query-agnostic notes, or QA pairs. Claims and notes include video identifiers and timestamps when available. QA pairs include the source videos used to produce the answer.

G.2 BULLET Generation

The bullet-point generator renders selected evidence items directly as a numbered list of findings with inline citations. This variant does not invoke a generation model and is intended as a conservative evidence-presentation baseline.

G.3 Single-Prompt Article Generation

The single-prompt article generator concatenates the evidence items for a query into a single prompt and generates a coherent article with inline citations. To reduce context length and memory failures, evidence sets larger than 25 items are truncated to the top 25 by confidence score.

G.4 GINGER Article Generation

The GINGER generator decomposes article generation into facet clustering, cluster ranking, per-cluster summarization, and fluency enhancement. Since the information extraction stage already produces atomic evidence units, the pipeline begins from extracted notes, claims, or QA pairs rather than running a separate nugget-detection stage.

H Appendix: RLM Controller Implementation

This appendix documents the tool API, memory schema, and prompts used by the RLM controller (see section 6 in the main text). The goal is to make the RLM-side of our submission directly reproducible. Figure 4 summarizes the resulting Think–Act–Observe control loop.

H.1 Tool API and Backing Modules

Table 18 lists all callable tool functions registered in the REPL namespace.

H.2 Memory Bank JSON Structure

```
{
  "findings":      [ " high-level
                      insights " ],
  "keywords":      { "<video_id>": [ "
                      keyword1", "keyword2" ] },
  "fact_table":    { "<video_id>": [
                      { "fact": "...", "timestamp": "10s
                      -15s", "source_tool": "
                      query_claims", "confidence":
                      0.8 }
                    ] },
  "selected_facts": [ "facts chosen by
                      llm_judge for the final report
                      " ],
  "videos":        { "<video_id>": { "
                      status": "processed", "
```

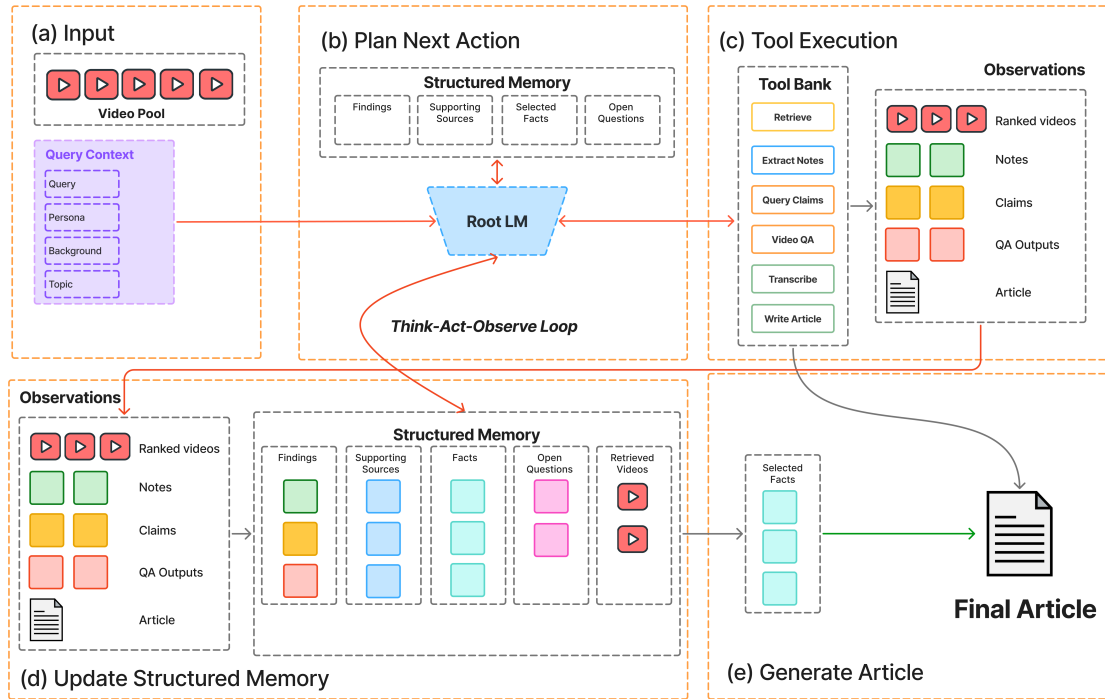


Figure 4: MARQUIS-RLM controller. The Root LM reads structured memory, plans the next action, executes one tool call in a persistent REPL environment, observes the result, and updates memory before continuing. Once sufficient evidence has been gathered and judged, selected facts are passed to the article-generation tool to produce the final cited article.

```
tools_used": [...], "path": "
...", "caption": "..."}
}
```

H.3 Examples of Root LM Behavior

Example 1 Final recheck.

```
[THINK] (6.5s):
The report has been generated but gaps remain. I
should verify one final critical issue - whether the
seat counts (Lib 120, CON 87) shown in zaFtBz84Kyk
were final or still live projections, since this
affects report accuracy. Let me verify this with
video_qa one more time.

[ACT]:
```repl
video_qa("zaFtBz84Kyk",
 "Was the seat count shown (Lib 120, CON 87)
the final result or a live projection? Clarify the
exact time shown and when results were finalized.")
```

In [42]:
1 video_qa("zaFtBz84Kyk", "Was the seat count shown
(Lib 120,
    CON 87) the final result or a live
projection? ...")

Out [42]:
'[video_qa] zaFtBz84Kyk - Q: Was the seat count
shown (Lib 120, CON 87) the final result or a live
projection? Clarify the exact time shown and when
```

results were finalized. A: The seat count shown (Lib 120, CON 87) was a live projection, not the final result, as the final declaration occurred later with the banner "LIB WIN" and the count updated to 124 for Liberals. confidence=1.0, timestamp=220-230s'

```
[OBSERVE]:
TOOL CALLS (1):
OK video_qa({'video_path': 'zaFtBz84Kyk', ...})
-> 82.7s
```

```
[JUDGE] iter 41 (model: qwen/qwen3.5-9b):
Reasoning 5 | Action 5 | Granularity 5 | Progress 5
Redundancy 5 | Think 5 | Code 5 | Output 5 | Tool 5
WEIGHTED TOTAL: 62.5/62.5 (100.0%)
```

Observation. Root LM re-engages video_qa to disambiguate a data point to prioritize correctness over completion.

Example 2 Vague Information Clarification.

```
[THINK] (3.4s):
I've watched all 17 videos but only queried 4 for
facts. I'm missing critical casualty and aid
distribution data. Video 1978302738418032640 caption
mentions "at least one is dead and several remain
missing" - I need to extract these numbers via
video_qa.
```

```
[ACT]:
```repl
video_qa("1978302738418032640","How many fatalities
and missing persons are there?What details about
```

Tool	Signature	Backing module
<i>Perception tools (multimodal extraction over raw video)</i>		
Caption	<i>video_caption(vid)</i>	local Qwen3.5-9B model
GeneralNotes	<i>general_notes(vid)</i>	<a href="#">subsection 4.1</a> general note extraction
QueryClaims	<i>query_claims(vid)</i>	<a href="#">subsection 4.2</a> query-conditioned claim extraction
<i>Targeted-query tools</i>		
VideoQA	<i>video_qa(vid, question)</i>	<a href="#">subsection 4.3</a> multimodal QA pipeline
Transcribe	<i>transcribe(vid)</i>	local Whisper Model
RetrieveChunks	<i>retrieve_chunks(vid)</i>	<a href="#">section 3</a> OmniEmbed retriever, lowered to 20 s chunk level
<i>Generation tool</i>		
WriteReport	<i>write_report(facts)</i>	<a href="#">section 5</a> GINGER-based pipeline
<i>Memory operators</i>		
memory_summary	<i>memory_summary()</i>	compact memory snapshot per iteration
print_memory	<i>print_memory(slot=None)</i>	full JSON dump of one or all slots
add_keyword	<i>add_keyword(vid, kw)</i>	tag a video with a keyword
search_by_keyword	<i>search_by_keyword(kw)</i>	find items in memory bank
remove_fact	<i>remove_fact(vid, idx)</i>	delete a single fact
clear_facts	<i>clear_facts(vid=None)</i>	clear all facts for one or all videos
<i>Memory operators</i>		
Think	<i>llm_think()</i>	fact_table → findings (inferred by LLM)
Judge	<i>llm_judge()</i>	fact_table → selected_facts (for report generation)

Table 18: Tools and memory operators registered in the RLM REPL namespace.

```

rescue operations?")
...

In [27]:
1 video_qa("1978302738418032640", "How many
fatalities and missing persons are there? ...")

Out [27]:
'["video_qa"] 1978302738418032640 - Q: How many
fatalities and missing persons are there? What details
about rescue operations?
A: At least one fatality and several missing; over
50 rescued via helicopter.
confidence=1.0, timestamp=8-15s'

[OBSERVE]:
TOOL CALLS (1):
OK video_qa({'video_path': '1978302738418032640',
...}) -> 22.2s

[JUDGE] iter 26 (model: qwen/qwen3.5-9b):
Reasoning 5 | Action 5 | Granularity 5 | Progress 5
Redundancy 5 | Think 5 | Code 5 | Output 5 | Tool 5
WEIGHTED TOTAL: 62.5/62.5 (100.0%)

```

**Observation.** Root LM identified the information from the caption was vague and decide to call the QA tool to get a more precise answer.

### Example 3 Cross-modal Conflict Resolution.

```

[THINK] (4.1s):
I need to resolve the major conflict - seat counts
vary between r2P6c5tzvM8 (LIB 152, CON 120) and
zaFtBz84KyK (LIB 119-120, CON 84-87). r2P6c5tzvM8
appears more authoritative as CTV News official report
. I should use transcribe to get the final declared
results from both CTV videos to reconcile this
conflict.

[ACT]:
...repl
transcribe("r2P6c5tzvM8")
...

```

```

In [17]:
1 transcribe("r2P6c5tzvM8")

Out [17]:
"Transcript of r2P6c5tzvM8 (2325 chars):
If you're Canadian, keep watching. It's election
time. I've just requested that the Governor General
dissolve Parliament and call it an election, and she
has agreed. The campaign will last 36 days, and the
election will be on April 28th. There are 343 ridings
up for grabs across the country. ... Here's what the
House looks like right now with the Liberal minority
government. The Liberals have 152 seats. The
Conservatives have 120. The Bloc Quebecois have 33.
The NDP have 24. The Green Party has 2. Three MPs are
sitting as independents, and four seats are vacant.
For a majority government, a party needs to win at
least 172 seats, but a minority government can be
formed with fewer than that. ..."

[OBSERVE]:
TOOL CALLS (1):
OK transcribe({'video_path': 'r2P6c5tzvM8'}) ->
32.2s

[JUDGE] iter 16 (model: qwen/qwen3.5-9b):
Reasoning 4 | Action 5 | Granularity 5 | Progress 4
Redundancy 5 | Think 5 | Code 5 | Output 5 | Tool 5
WEIGHTED TOTAL: 60.5/62.5 (96.8%)

```

**Observation.** When root gets conflicting visual information, it decides to call transcribe to use transcript evidence for cross-modal conflict resolution.

## H.4 Statistics

**Tool Usage.** As shown in [Table 19](#), average tool usage indicates a strongly adaptive exploration strategy. In practice, the agent schedules all tools in 14 of 19 queries. In practice, it uses Caption tool ( $10 \pm 3$ ) mainly as a relevance filter, and relies more

	Tool Calls (Average)						
	Capt.	Claims	Notes	QA	Trans.	Retr.	Total
Q1	6	6	4	6	4	1	28
Q2	6	6	4	6	2	2	26
Q3	5	6	5	4	2	2	24
Q4	5	5	2	2	2	0	15
Q5	8	6	2	2	1	1	21
Q6	7	10	4	4	2	2	29
Q7	11	12	3	4	1	1	32
Q8	10	10	4	2	2	1	28
Q9	7	7	1	1	0	0	16
Q10	17	5	3	4	0	1	30
Q11	18	10	2	2	0	0	32
Q12	11	8	7	2	3	2	33
Q13	10	5	6	2	2	0	24
Q14	10	7	3	2	1	1	24
Q15	10	8	4	3	2	1	28
Q16	10	10	4	1	1	1	26
Q17	11	12	6	2	1	0	32
Q18	10	10	2	1	1	2	26
Q19	10	10	1	1	1	1	24
Avg.	10 $\pm$ 3	8 $\pm$ 2	4 $\pm$ 2	3 $\pm$ 2	2 $\pm$ 1	1 $\pm$ 1	26 $\pm$ 5

Table 19: Statistics of tool calls per query (averaged over two runs).

on targeted extraction with QueryClaim tool( $8 \pm 2$ ) than on broader summarization with General Notes tool ( $4 \pm 2$ ). Transcribe tool is used only as a supplement, while Retrieval tool is rarely needed ( $1 \pm 1$ ), since most videos are short ( $< 60$ s) and do not require fine-grained temporal localization.

**Runtime Performance.** The evaluation by LLM-as-a-judge indicates strong behavioral quality ( $92 \pm 2\%$  on average), with near-perfect Output Waste and Code Minimality scores as reported in Table 20. The average wall time per query is  $36 \pm 12$  minutes, with most of the runtime spent on tool execution rather than Root LM inference.

**Root LM Context Window Usage** As detailed in Table 21, Root LM use an average of only 33% of its 32K context window, avoiding frequent truncation. The context growth scales linearly to roughly 1.3% per iteration (from 8% at Iteration 0 to 61% at Iteration 40). This stable progression indicates that the accumulation of both the Memory Bank and history is controllable, therefore preventing explosive token consumption in later iterations.

## I Appendix: Prompt Templates

This appendix collects all prompt templates used by the system. Prompts are grouped by the method component that uses them. Implementation details for each method are provided in Appendices C–H.

	Quality		Latency		
	Judge (%)	Facts/Iter	Wall (min)	Tokens (K)	Root LLM (s)
Q1	88	0.74	46	595	382
Q2	92	0.57	47	510	326
Q3	86	0.90	30	516	311
Q4	94	1.85	24	329	216
Q5	95	1.11	36	405	278
Q6	92	0.37	49	476	342
Q7	92	1.08	42	493	416
Q8	89	1.72	48	563	409
Q9	91	1.39	16	175	199
Q10	93	0.99	26	418	357
Q11	93	1.56	22	430	300
Q12	93	1.28	22	472	481
Q13	92	1.00	22	271	346
Q14	92	0.93	36	457	366
Q15	89	1.21	45	470	351
Q16	92	0.86	48	613	342
Q17	90	0.81	54	419	298
Q18	94	0.91	37	378	235
Q19	95	1.26	26	329	244
Avg.	92 $\pm$ 2	1.1 $\pm$ 0.4	36 $\pm$ 12	438 $\pm$ 109	326 $\pm$ 72

Table 20: Statistics of behavior quality and query latency, averaged over two runs.

### I.1 Retrieval Query Expansion Prompt

**Sub-query decomposition prompt.** The decomposition prompt consumes the structured fields of a MAGMaR query and emits a JSON array of fine-grained search phrases. It is used with Qwen3.5-72B and thinking disabled.

```
You are a research decomposition specialist.
Your task is to take a user's query
and break it down into an exhaustive set of
searchable sub-queries - short
phrases or keyword combinations that could be
entered into a search engine or
database to retrieve all the information
needed to fully answer the original query.
You will receive the following inputs:
- Title: <TITLE>
- Language: <LANGUAGE>
- Persona: <PERSONA_TITLE>
- Background: <BACKGROUND>
- Query: <QUERY>
Decomposition Rules:
1. Coverage: Extract every distinct piece
of information the user is asking for. Do
not merge separate information needs into one
sub-query.
2. Granularity: Each sub-query should
target ONE specific, retrievable piece of
information. Prefer atomic queries over
compound ones.
3. Implicit needs: Go beyond what is
explicitly stated. Based on the background
and persona_title, infer what additional
information the user would likely need but
did not explicitly ask for.
```

Metric	Value
<i>Per-Iteration Token Consumption</i>	
Total tokens	11131
Prompt (context)	10689 $\pm$ 6434 (96%)
Completion	442 $\pm$ 415 (4%)
Reasoning	328
Output	114
<i>Context Window Utilization (32K limit)</i>	
Average usage	33% $\pm$ 20%
Maximum usage	97%
Iterations >80%	25 (1.7%)
<i>Context Growth Over Iterations</i>	
Iteration 0	2,582 (8%)
Iteration 20	9,983 (31%)
Iteration 40	20,026 (61%)

Table 21: Root LM context-window usage, averaged over two runs.

4. Search-friendly format: Each sub-query should be phrased as a concise search phrase, typically 3–10 words, not a full sentence or question.
  5. Context anchoring: Each sub-query should include enough context to be independently searchable without ambiguity.
  6. Source-awareness: If the user requests source information, generate sub-queries targeting official sources, methodologies, and data provenance.
  7. Dimensional expansion: Consider additional perspectives or breakdowns by time, place, category, cause, mechanism, or comparison only when they add value.
  8. No redundancy: Each sub-query must be meaningfully distinct.
  9. Language: Always generate sub-queries in English.
  10. Generate between 10 and 25 sub-queries.
  11. Do not mechanically prepend the full topic title to every sub-query.
  12. Focus on the specific information being sought, not on repeating the topic name.
- Return ONLY a JSON array of strings. No explanation, no markdown, no code blocks.

**General note extraction prompt.** The general-note prompt is query-agnostic but not fully context-free: it includes the source topic and video identifier together with an evidence-first instruction block.

You are extracting observation notes directly from a raw video.  
Video context:  
- topic: <TOPIC>  
- video\_id: <VIDEO\_ID>  
- timestamp\_span: <TIMESTAMP\_SPAN\_OR\_NULL>  
Rules:  
- Record only directly observable content.  
- No inference, speculation, causality, or cross-video synthesis.

- Capture OCR (on-screen text), events, and visible scene details.
- One note per atomic visible, audible, or textual fact.
- Use modality ‘visual’ for scene content, ‘ocr’ for on-screen text, and ‘audio’ for transcript or speech.
- Use the provided timestamp span for each note when no narrower timestamp is available.
- If there is no usable evidence, return an empty notes list.

Output strict JSON only.  
No markdown, no code fences, no explanation, no extra keys outside the schema.  
Expected shape:

```
{
 "notes": [
 {
 "text": "...",
 "modality": "visual",
 "timestamp": [0.0, 1.0]
 }
]
}
```

### Query-conditioned claim extraction prompt.

The single-query claim-extraction prompt conditions on the query text together with persona, background, topic, and video identity.

You are extracting query-relevant claims directly from a raw video.  
Query context:  
- query\_id: <QUERY\_ID>  
- topic: <TOPIC>  
- persona\_title: <PERSONA\_TITLE>  
- background: <BACKGROUND>  
- query: <QUERY\_TEXT>  
- video\_id: <VIDEO\_ID>  
Rules:  
- Extract up to <PER\_VIDEO\_TARGET> claims relevant to the query from this video.  
- Claims must be directly supported by observable video content.  
- Avoid generic scene summary unless it directly serves the query.  
- Avoid duplicates and paraphrases.  
- If the video does not contain evidence for the query, return an empty claims list.  
- source must be one of ‘video\_visual’, ‘video\_text’, or ‘transcript’.  
- timestamp must be [start, end].  
- confidence must be a float between 0 and 1.  
Output strict JSON only.  
No markdown, no code fences, no explanation, no extra keys outside the schema.  
Expected shape:

```
{
 "claims": [
 {
 "claim": "...",
 "confidence": 0.85,
 "evidence": "...",
 "source": "video_visual",
 "timestamp": [0.0, 1.0]
 }
]
}
```

```
]
}
```

**Decompose Query into Questions.** Expand single query into a series of question-answer pairs based on a provided title, language, persona, and background.

You are a research decomposition specialist. Your task is to take a user's query and break it down into an exhaustive set of searchable research questions – complete questions that could be used to retrieve all the information needed to fully answer the original query. You will receive the following inputs:

- Title: {title}
- Language: {language}
- Persona: {persona\_title}
- Background: {background}
- Query: {query}

Decomposition Rules:

1. Coverage: Extract every distinct piece of information the user is asking for. Do not merge separate information needs into one question. If the query asks for multiple related but distinct data points, each one should become its own question.
2. Granularity: Each question should target ONE specific, retrievable piece of information. Prefer atomic questions over compound ones.
3. Implicit needs: Go beyond what is explicitly stated. Based on the background and persona\_title, infer what additional information the user would likely need but did not explicitly ask for. Consider what a professional in that role would typically require to produce complete, high-quality work on this topic.
4. Search-friendly format: Each sub-query must be written as a concise, well-formed question that could plausibly be entered into a search engine or research database.
5. Context anchoring: Each question should include enough context (e.g., specific names, dates, locations, technical terms) to be independently searchable without ambiguity.
6. Source-awareness: If the user requests source information or credibility indicators, generate questions specifically targeting official sources, methodologies, and data provenance.
7. Dimensional expansion: For each core information need identified, consider whether the user would benefit from additional perspectives or breakdowns. Ask yourself: can this information be meaningfully decomposed further by time, place, category, cause, mechanism, comparison, or any other axis that is natural and relevant to the topic? Only expand along dimensions that genuinely add value given the query's subject matter and the user's background.
8. No redundancy: Each question must be meaningfully distinct. Do not produce near-duplicates that would return the same

search results.

9. Language: Always generate questions in English, regardless of the language field in the input.

10. Quantity: Generate between 10 and 25 questions. Focus on quality and relevance over quantity.

11. Avoid mechanical repetition: Do not mechanically prepend the full topic title to every question. Each question should contain only the context necessary for an effective search.

12. Focus on information needs: Focus on the specific information being sought rather than repeating the topic name unnecessarily.

Return ONLY a JSON array of strings. No explanation, no markdown, and no code blocks. For example, given a query about the 2025 Canadian federal election asking for seat counts and vote shares, good questions would be:

```
["What was the total number of seats won by each political party in the 2025 Canadian federal election?", "What percentage of the national popular vote did each major party receive in the 2025 Canadian federal election?", "How many seats did each party gain or lose compared with the 2021 Canadian federal election?", "What official datasets published by Elections Canada contain vote totals and seat counts for the 2025 federal election?", "What demographic voting patterns were observed in the 2025 Canadian federal election?"]
```

NOT:

```
["What happened in the 2025 Canadian federal election?", "What were the results of the 2025 Canadian federal election?", "What information is available about the 2025 Canadian federal election?"]
```

JSON array:

**Question Answering.** Qwen 3.5 produces an answer to the question based on the down-sampled video and transcript.

Question: question

Answer concisely using the video and transcript. Answer to the best of your abilities. If you can't answer all of the question, answer the parts that you can (Ex. if asked about Liberal and Conservative vote counts, but only have the Liberal vote counts, return those). If you have no information about anything related to the question, return "I don't know". Never say the event hasn't taken place.

Return ONLY the final factual answer.

**Combined Answers.** Combines all the answers generated independently from each relevant video in the top k most similar.

You are given extracted answers from videos. These answers are factual and must be used.  
 Question: {subquery}  
 Extracted Answers (treat as ground truth): {valid\_answers}  
 Combine them into a single answer.  
 Do NOT use prior knowledge. Do NOT say the event has not taken place. Only use the provided answers.  
 If you receive conflicting information, make a best guess NOT on prior knowledge but based on the nature of the question (Ex. for a question about how many seats a party has one, its reasonable to assume the largest number is the most recent seat count)  
 Return only the final answer. You might not be able to answer it fully, and that's okay. Answer what you can and then say specifically what information is unknown.

**Generate Follow Up Question.** Generates the follow up question in the iterative QA system based on entire past context.

You are refining a research question based on prior answers.  
 Context: context  
 Generate ONE new question that: - extracts new information not yet covered - is more specific or differently framed  
 If no meaningful new question can be formed, output: NONE  
 Return ONLY the question or NONE.

**Qwen 3.5 calibration prompt.** The prompted Qwen 3.5 backend receives the full source video together with one extracted artifact and is instructed to return a scalar probability in a constrained answer format.

To help you make more accurate and consistent judgments, here is an expanded explanation of how to interpret and assign support percentages.  
 These examples are designed to cover a range of real-world cases you may encounter in the annotation task.  
 100% - /Full and unambiguous support:  
 The video clearly shows the exact event described in the claim. There is no need for guessing or interpretation.  
 80-100% - Almost complete support:  
 The main content in the claim is shown, but there may be minor ambiguity in location, identity, or completeness. The overall claims are supported by the video.  
 60-80% - Strong partial support:  
 The video strongly suggests the claim is true, but some critical details may be missing, obscured, or ambiguous, limiting the ability to confirm the claim with certainty. The video gives strong but not definitive support.  
 40-60% - Moderate partial support:

There is some alignment with the claim, but large portions are either missing, unclear, or open to interpretation. While the footage may point in the same general direction as the claim, it lacks the clarity or completeness needed for confident verification.  
 20-40% - Minimal weak support:  
 There are small visual or audio cues that could hint at the claim, but they are insufficient to be confident.  
 0-20% - Very weak or speculative support:  
 There may be the slightest indirect reference, such as a related object or setting, but nothing concrete happens.  
 0% - No support or contradiction:  
 The video does not relate to the claim at all, or it directly shows something opposite.  
 Based on the provided video and text, evaluate the probability that the text is true.  
 Your answer must be a decimal number between 0 and 1, and you must strictly follow the format below:  
 <answer>probability\_value</answer>  
 Where probability\_value is the result you calculate.  
 The text to evaluate is:  
 <ARTIFACT\_TEXT>

Malformed outputs trigger a stricter retry prompt that preserves the same task content while requiring a single answer in the exact form  
 <answer>0.73</answer>.

**Baseline** The baseline approach feeds all query-conditioned claims into a single LLM prompt and generates the complete report in one forward pass.

You are a report writing assistant. Your task is to synthesize a set of claims extracted from multiple videos into a single, fluent, well-organized report that answers the given query.  
 ## Instructions:  
 1. Read all the claims below carefully. Each claim was extracted from a specific video and has a timestamp.  
 2. Group related claims together logically (e.g., by sub-topic or chronological order).  
 3. Write a coherent, well-structured report that covers all the key information from the claims.  
 4. For EVERY piece of information in your report, include an inline citation in the format [video\_id, timestamp\_start-timestamp\_end].  
 5. If multiple claims from different videos support the same point, cite all relevant sources.  
 6. Remove redundant information – if multiple claims say the same thing, mention it once and cite all sources.  
 7. The report should be fluent and readable, not a list of bullet points.  
 8. Keep the report concise but comprehensive (aim for 200-400 words).

```
Query/Topic: {topic}
Claims:
{claims_text}
Report:
```

**GINGER clustering prompt.** The model receives all claims for a query and is instructed to partition them into thematic facet clusters, returning a labeled JSON partition of the claim set.

```
You are an information analyst. Given a set of claims about a topic extracted from videos, group them into distinct facet clusters. Each cluster should represent a different sub-topic or aspect of the main topic.
Instructions:
1. Read all claims carefully.
2. Group them into clusters based on their sub-topic/facet (e.g., "casualties", "rescue efforts", "damage assessment", "government response", etc.).
3. Each claim should belong to exactly one cluster.
4. Give each cluster a short, descriptive label.
5. Output your result as a JSON object with the following format:
{
 "clusters": [
 {
 "label": "Short descriptive label for this facet",
 "claim_ids": ["qc-10-xxx-000", "qc-10-xxx-001"]
 },
 ...
]
}
Only output the JSON object, no other text.
Topic: {topic}
Claims:
{claims_text}
```

**GINGER ranking prompt.** The model receives the labeled clusters and is instructed to rank them by relevance to the query topic, returning an ordered JSON array of cluster labels.

```
You are a relevance assessor. Given a query/topic and a list of facet clusters (each containing grouped claims from videos), rank the clusters from most to least relevant to the query.
Instructions:
1. Consider which facets are most important for answering/addressing the query topic.
2. Rank all clusters from most relevant to least relevant.
3. Output a JSON array of cluster labels in order from most to least relevant:
{
 "ranked_labels": ["most relevant label", "second most relevant", ...]
}
```

```
Only output the JSON object, no other text.
Topic: {topic}
Clusters:
{clusters_text}
```

**GINGER summarization prompt.** The model receives the claims within a single cluster and is instructed to condense them into one cited sentence of at most 40 words, preserving inline citations anchored to the supporting evidence.

```
You are a concise summarizer. Summarize the following cluster of claims into a SINGLE sentence (maximum 40 words). The sentence must:
1. Capture the key information from all claims in this cluster.
2. Include inline citations in the format [video_id, timestamp] for every fact mentioned.
3. Be factual – only include information present in the claims.
Cluster: {cluster_label}
Claims in this cluster:
{cluster_claims_text}
One-sentence summary:
```

**GINGER fluency prompt.** The model receives the concatenated one-sentence cluster summaries and is instructed to rewrite them into a coherent 200–400-word prose report without adding new information or removing any citations.

```
You are an editor. Below is a report composed of individual summary sentences about the topic "{topic}". Your task is to rewrite it into a smooth, fluent, well-organized report.
Rules:
1. Do NOT add any new information that is not in the summaries below.
2. Do NOT remove any information or citations from the summaries.
3. Keep ALL inline citations in the format [video_id, timestamp].
4. Improve transitions between sentences for better readability.
5. You may reorder sentences for better logical flow.
6. Keep the report concise (200-400 words).
Draft report (concatenated summaries):
{draft_report}
Final polished report:
```

**MARQUIS-RLM REPL system prompt.**

```
You answer queries using an interactive Python REPL, called iteratively until you submit a final answer.
THINK-ACT-OBSERVE LOOP:
Each iteration: THINK (brief reasoning), ACT (one code block), OBSERVE the output.
THINK phase: READ the memory snapshot below – it shows your findings (global knowledge)
```

and per-video facts. Base your next action on what you ALREADY KNOW, not assumptions.

{pacing}

ENVIRONMENT:

- context['task'], context['video\_pool'], context['tools'] are read-only.
- 'memory' is a persistent dict (survives compaction).
- Tools are pre-loaded as plain Python functions; call them directly.

FORMAT: THINK (2-4 sentences), then ONE "repl" code block (1-5 lines, ONE tool call). NO for-loops over videos.

FINAL ANSWER: report = write\_report(memory['selected\_facts']), then FINAL\_VAR(report) outside the code block.

### MARQUIS-RLM Root LM Think prompt.

TASK: {query\_text}

CURRENT FINDINGS: {findings\_str}

FACT TABLE SUMMARY: {fact\_summary}

VIDEO STATUS: {video\_status}

You are the analytical brain. Based on all facts collected so far:

1. NEW\_FINDINGS: List any new high-level findings (one sentence each) not already in CURRENT FINDINGS. If a new fact CONTRADICTS an existing finding, say CONFLICT: <existing> vs <new>.
2. UPDATED\_FINDINGS: Output the complete updated findings list (old + new, deduplicated). One finding per line, prefixed with '- '.
3. NEXT\_STEPS: What should the agent do next? Be specific: which video, which tool, which question. Be concise.

### MARQUIS-RLM Root LM Judge prompt.

TASK: {query\_text}

FINDINGS (root's current understanding): {findings\_str}

FACT TABLE ({n} facts): {fact\_lines}

You are a strict quality judge. Review ALL facts above for the task.

1. ITEM REVIEW: For each fact (F#0, F#1, ...), give a verdict.  
BE CONSERVATIVE - only REMOVE if clearly irrelevant or duplicate. When in doubt, KEEP.  
KEEP - useful, specific, or even mildly relevant (default)  
REMOVE - clearly irrelevant or duplicate of another listed fact  
REWRITE - needs more detail or has a missing timestamp (flag, do NOT drop)  
Format: F#0: KEEP / F#3: REMOVE (dup of F#1) / F#5: REWRITE (missing timestamp)
2. SELECTED: Pick the 10-40 BEST facts for a comprehensive report (prefer MORE coverage). List their IDs: SELECTED: F#0, F#2, F#7, ...

3. MISSING TIMESTAMPS: List facts that are useful but lack timestamps; suggest video\_qa queries to resolve them.
4. GAPS: What information is still missing for a thorough report?
5. READY: Can we write a good report now? (yes / no / almost)  
Be specific and concise.

### MARQUIS-RLM LLM-as-a judge prompt (behavior-level).

You are evaluating an AI agent's performance on iteration {iteration}/{max\_iter}.

TASK: {query}

MEMORY STATE BEFORE: {mem\_before}

THINK: {think\_text}

ACT: {code}

OBSERVE: {observe}

MEMORY STATE AFTER: {mem\_after}

Rate each dimension 1-5 with ONE sentence justification.

## Core dimensions:

1. Reasoning (1-5): Did THINK show sound reasoning based on memory?
2. Action (1-5): Was the chosen action relevant and logical?
3. Granularity (1-5): One focused step, or too much at once?
4. Progress (1-5): Did this iteration meaningfully advance the task?

## Efficiency breakdown (5 sub-scores):

- 5a. Eff\_Redundancy (1-5) - avoided repeating a tool call?
- 5b. Eff\_Think\_Conciseness (1-5) - THINK tight and non-repetitive?
- 5c. Eff\_Code\_Minimality (1-5) - minimal code for its purpose?
- 5d. Eff\_Output\_Waste (1-5) - avoided producing useless output?
- 5e. Eff\_Tool\_Choice (1-5) - most cost-effective tool for this sub-goal?

Format EXACTLY: one line per dimension as 'Name: <score> - <reason>', then 'TOTAL: <sum>/45'.